THE CATEGORY OF ADEQUATE MODULES IS ABELIAN AND ITS INCLUSION IS EXACT

THE STACKS PROJECT AUTHORS

Editorial note. All delivered mathematical text is the author's own, copied from the pinned Stacks Project chapter file adequate.tex; no statement or proof step was rewritten, reordered or removed.

Editorial note. The counted spans carry navigation stubs for the references leaving them; no bibliography citation occurs in the spans.

Editorial note. The preamble span points at the chapter's own preamble.tex shipped with the frozen source, used verbatim as staged.

Allowed standard prerequisites (editorial).

- Kernels and cokernels of maps of adequate functors are adequate. Same chapter, Lemmas lemma-kernel-adequate and lemma-cokernel-adequate, invoked at proof lines 2053-2056. Give adequacy of the kernel and cokernel in the abelian category of functors.
- Adequacy is preserved by sheafification, and the associated functor construction. Same chapter, Lemma lemma-sheafification-adequate, invoked at proof lines 2060-2064. Transfers adequacy from the presheaf cokernel to the cokernel.
- Vanishing of H^1 for adequate modules on affine schemes, and exactness of the associated functor sequence. Same chapter, Lemma lemma-same-cohomology-adequate, invoked at proof lines 2075-2077. Makes the sequence of functors associated to a short exact sequence exact.
- Adequacy is local, and an extension of adequate functors is adequate. Same chapter, Lemmas lemma-adequate-local and lemma-extension-adequate, invoked at proof lines 2078-2081. Gives item (3).
- Colimits of adequate functors and their adequacy. Same chapter, Lemma lemma-colimit-adequate, invoked at proof lines 2084-2099. Gives item (4).

Source reference. Original lemma 06VI: sources/stacks-adequate/pinned-original-source.tex, source lines 1707–1707. Pointer outside the counted span, invoked once for context; the load-bearing argument of the counted statement is the author's proof reproduced in full in the proof span.

Source reference. Original lemma 06V2: sources/stacks-adequate/pinned-original-source.tex, source lines 684–684. Pointer outside the counted span, invoked once for context; the load-bearing argument of the counted statement is the author's proof reproduced in full in the proof span.

Source reference. Original lemma 06V4: sources/stacks-adequate/pinned-original-source.tex, source lines 734–734. Pointer outside the counted span, invoked once

Date: stacks-project:adequate.tex:local-snapshot-20261012; GNU FDL 1.2 or later.

for context; the load-bearing argument of the counted statement is the author's proof reproduced in full in the proof span.

Source reference. Original lemma 06V8: sources/stacks-adequate/pinned-original-source.tex, source lines 910–910. Pointer outside the counted span, invoked once for context; the load-bearing argument of the counted statement is the author's proof reproduced in full in the proof span.

Source reference. Original lemma 06V3: sources/stacks-adequate/pinned-original-source.tex, source lines 709–709. Pointer outside the counted span, invoked once for context; the load-bearing argument of the counted statement is the author's proof reproduced in full in the proof span.

Source reference. Original lemma 06VM: sources/stacks-adequate/pinned-original-source.tex, source lines 1896–1896. Pointer outside the counted span, invoked once for context; the load-bearing argument of the counted statement is the author's proof reproduced in full in the proof span.

Source reference. Original lemma 06VP: sources/stacks-adequate/pinned-original-source.tex, source lines 1985–1985. Pointer outside the counted span, invoked once for context; the load-bearing argument of the counted statement is the author's proof reproduced in full in the proof span.

Lemma 0.1. *Let S be a scheme.*

1. *The category $Adeq(\mathcal{O})$ is abelian.*
2. *The functor $Adeq(\mathcal{O}) \to Mod((Sch/S)_\tau, \mathcal{O})$ is exact.*
3. *If $0 \to \mathcal{F}_1 \to \mathcal{F}_2 \to \mathcal{F}_3 \to 0$ is a short exact sequence of $\mathcal{O}$-modules and $\mathcal{F}_1$ and $\mathcal{F}_3$ are adequate, then $\mathcal{F}_2$ is adequate.*
4. *The category $Adeq(\mathcal{O})$ has colimits and $Adeq(\mathcal{O}) \to Mod((Sch/S)_\tau, \mathcal{O})$ commutes with them.*

Proof. Let $\varphi : \mathcal{F} \to \mathcal{G}$ be a map of adequate $\mathcal{O}$-modules. To prove (1) and (2) it suffices to show that $\mathcal{K} = \mathrm{Ker}(\varphi)$ and $\mathcal{Q} = \mathrm{Coker}(\varphi)$ computed in $Mod((Sch/S)_\tau, \mathcal{O})$ are adequate. Let $U = \mathrm{Spec}(A)$ be an affine scheme over S. Let $F = F_{\mathcal{F},A}$ and $G = F_{\mathcal{G},A}$. By Lemmas 06V3 and 06V2 the kernel K and cokernel Q of the induced map $F \to G$ are adequate functors. Because the kernel is computed on the level of presheaves, we see that $K = F_{\mathcal{K},A}$ and we conclude $\mathcal{K}$ is adequate. To prove the result for the cokernel, denote $\mathcal{Q}'$ the presheaf cokernel of φ. Then $Q = F_{\mathcal{Q}',A}$ and $\mathcal{Q} = (\mathcal{Q}')^\#$. Hence $\mathcal{Q}$ is adequate by Lemma 06VP.

Let $0 \to \mathcal{F}_1 \to \mathcal{F}_2 \to \mathcal{F}_3 \to 0$ is a short exact sequence of $\mathcal{O}$-modules and $\mathcal{F}_1$ and $\mathcal{F}_3$ are adequate. Let $U = \mathrm{Spec}(A)$ be an affine scheme over S. Let $F_i = F_{\mathcal{F}_i,A}$. The sequence of functors

$$0 \to F_1 \to F_2 \to F_3 \to 0$$

is exact, because for $V = \mathrm{Spec}(B)$ affine over U we have $H^1(V, \mathcal{F}_1) = 0$ by Lemma 06VM. Since F_1 and F_3 are adequate functors by Lemma 06VI we see that F_2 is adequate by Lemma 06V8. Thus $\mathcal{F}_2$ is adequate.

Let $\mathcal{I} \to Adeq(\mathcal{O})$, $i \mapsto \mathcal{F}_i$ be a diagram. Denote $\mathcal{F} = \mathrm{colim}_i \mathcal{F}_i$ the colimit computed in $Mod((Sch/S)_\tau, \mathcal{O})$. To prove (4) it suffices to show that $\mathcal{F}$ is adequate. Let $\mathcal{F}' = \mathrm{colim}_i \mathcal{F}_i$ be the colimit computed in presheaves of $\mathcal{O}$-modules. Then $\mathcal{F} = (\mathcal{F}')^\#$. Let $U = \mathrm{Spec}(A)$ be an affine scheme over S. Let $F_i = F_{\mathcal{F}_i,A}$. By

Lemma 06V4 the functor $\operatorname{colim}_i F_i = F_{\mathcal{F}',A}$ is adequate. Lemma 06VP shows that $\mathcal{F}$ is adequate. $\square$